Radiant energy: Q [J] 光源发出的总能量.

Radiant Flux: $\Phi = \frac{dQ}{dt}$ [W] [lm] 可理解为光源总共的辐射功率

Radiant Intensity: $I(w) = \frac{d\Phi}{dw}$ $[\frac{W}{sr}]$ [cd] 表示点光源每单位立体角总共辐射功率。对于点光源它非常重要。

立体角: $w = \frac{A}{r^2}$ [sr]

$dA = (rd\theta)(r\sin\theta d\phi)$

$= r^2 \sin\theta d\theta d\phi$

$dw = \frac{dA}{r^2} = \sin\theta d\theta d\phi$

$\Omega = \int_{S^2} dw = \int_0^{2\pi}\int_0^{\pi} \sin\theta d\theta d\phi = 4\pi$ [sr]

$\Phi = \int_{S^2} d\Phi = \int_{S^2} I(w) dw = 4\pi I$

$\Rightarrow I = \frac{\Phi}{4\pi}$

Irradiance: $E(p) = \frac{d\Phi(p)}{dA\cos\theta}$ $[\frac{W}{m^2}]$ $[\frac{lm}{m^2} = lux]$ 辐照度描述一个表面实际接收到的光能密度.

$E' = \frac{\Phi}{4\pi r^2} = \frac{E}{r^2}$

Radiance: $L(p,w) = \frac{d^2\Phi(p,w)}{dw dA\cos\theta}$ $[\frac{W}{sr\,m^2}]$ $[\frac{cd}{m^2} = nit]$

$L(p,w) = \frac{dE(p)}{dw\cos\theta}$ 辐亮度：描述某位置，某个向上的光的强度密度。

入射角度

$L(p,w) = \frac{dI(p,w)}{dA\cos\theta}$

出射角度.

$dE(p,w) = L_i(p,w)\cos\theta dw$

$E(p) = \int_{H^2} L_i(p,w)\cos\theta d\theta$.

扫描全能王 创建

Rendering Equation

$$L_o(p,\omega_o) = L_e(p,\omega_o) + \int_{\Omega} f_r(p,\omega_i,\omega_o) L_i(p,\omega_i)(n\cdot\omega_i)^+ d\omega_i$$

Pitferential Irradiance incoming: $dE(\omega_i) = L(\omega_i)\cos\theta_i d\omega_i$

Differential radiance exiting: $dL_r(\omega_r)$

$$f_r(\omega_i \to \omega_r) = \frac{dL_r(\omega_r)}{dE_i(\omega_i)} = \frac{dL_r d\omega_r}{[L(\omega_i)\cos\theta_i d\omega_i]\left[\frac{1}{sr}\right]}$$

$$h = \frac{l+v}{\|l+v\|}$$

$\max(0, n\cdot l)$

扫描全能王 创建

Cook-Torrance 微表面 BRDF.

$f_r = f_{diffuse} + f_{specular}$.

漫反射项　镜面反射项.

$$f_{specular} = \frac{D(\vec{n},\vec{h})F(\vec{v},\vec{h})G(\vec{n},\vec{l},\vec{v})}{4(\vec{n}\cdot\vec{l})(\vec{n}\cdot\vec{v})}$$

D: Normal Distribution Function

F: Fresnel Term

G: Geometry

F: Schlick 近似.

$F = F_0 + (F_{90} - F_0)[1 - \max(0, \vec{v}\cdot\vec{h})]^5$.

F_{90} 通常取 1.

$F = F_0 + (1 - F_0)[1 - \max\{0, \vec{v}\cdot\vec{h}\}]^5$

F_0: 垂直入射时的反射率.

$$F_0 = \left(\frac{1-\eta_i}{1+\eta_i}\right)^2 = \left(\frac{1-1.5}{1+1.5}\right)^2 = 0.04$$

M-R-T 作法中金属通常用 base color 作为 F_0.

D: GGX/Trowbridge-Reitz

$$D_{GGX} = \frac{\alpha^2}{\pi[(\max\{\vec{n}\cdot\vec{h}, 0\})^2(\alpha^2-1)+1]^2}$$

$\alpha = r^2$.

G: Smith 阴影遮蔽

$G(N, L, V) = G(N, L)\,G(N, V)$

$$\Lambda(\vec{n},\vec{w}) = \frac{\sqrt{1+\alpha^2\frac{(1-(\vec{n}\cdot\vec{w})^2)}{(\vec{n}\cdot\vec{w})}} - 1}{2}$$

$$G(\vec{n},\vec{w}) = \frac{1}{1+\Lambda(\vec{n},\vec{w})}$$

G 项

$k = \frac{(r+1)^2}{8}$

$$G_1(V) = \frac{n\cdot V}{(n\cdot V)(1-k)+k}$$

$G(L, V, h) = G_1(L)\,G_1(V)$.

漫反射：

① Lambert 漫反射模型

$f_{diffuse} = \frac{c}{\pi}$.

② 金属度引入(m)

$f_{diffuse} = \frac{(1-m)c}{\pi}$.

③ 严格的能量守恒

$k_d = (1-F)(1-m)$

$f_{diffuse} = \frac{k_d c}{\pi} = \frac{(1-F)(1-m)c}{\pi}$

扫描全能王 创建

Metallic-Roughness 工作流.

① Base color.

C = BaseColorTexture × BaseColorFactor

② Metallic:

m = clamp(T^B_MR · m_factor, 0, 1)

③ Roughness

r = clamp(T^G_MR · r_factor, 0, 1)

G 中, α = r².

④ F0 如何得到:

F0 = 0.04(1 − m) + C·m

= mix(0.04, C, m)

m=0, F0 = 0.04,

m=1, F0 = C

直接光照

对于一个像素点来一个直接光源

$L_o = L_o^{direct} = (f_{diffuse} + f_{specular}) L_i \cdot (\vec{n} \cdot \vec{l})$

$= \left[\frac{(1-F)(1-m)C}{\pi} + \frac{DFG}{4(\vec{v}\cdot\vec{n})(\vec{l}\cdot\vec{n})} \right] \cdot L_i(\vec{n}\cdot\vec{l})$

扫描全能王 创建

间接光 IBL

$L_o = \int_{\Omega} L_i(l)(n \cdot l)\, dw$

$L_o^{IBL} = L_o^{diffuse} + L_o^{specular\ IBL}$

漫反射 IBL

$E(\vec{n}) = \int_{\Omega} L_i(l)(n \cdot l)\, dw$

对于 Lambert 漫反射：$L_o^{diffuse\ IBL} = \frac{c}{\pi} E(\vec{n})$

对于 PBR 模型：$L_o^{diffuse} \approx (1+m)\left(\frac{c}{\pi}\right)$ 预积分

镜面 IBL

$L_o^{specular} = \int_{\Omega} f_{specular}(l, v) L_i(l)(n \cdot l)\, dw$

$L_o^{specular\ IBL} \approx L_{prefiltered}(R, V)\left[F_0 A(\vec{n} \cdot \vec{v}, r) + B(\vec{n} \cdot \vec{v}, r)\right]$

A_o：$A_{effective} = HS(A_{sample} - 1)$

$L_o^{specular} \approx A_{esp}$

扫描全能王 创建